LAB #7

Adv. CiC
Spring 2026

Agenda

1. Housekeeping
2. Databases
3. SQL
4. Lab 7 Exercise

Housekeeping

Due this past week

- Reading 7
- Lab 6

Upcoming

- Project pt. 4 - **3/12 @ 3pm**
- Lab 7 - **3/13 @ 1pm**

Announcements

1. Project evolution / regrades
2. Open Data Week

Questions

(on Lectures, Labs, or Readings)?

Databases

What is a database?

an **organized system that stores data** so it can be easily accessed and updated by users or applications

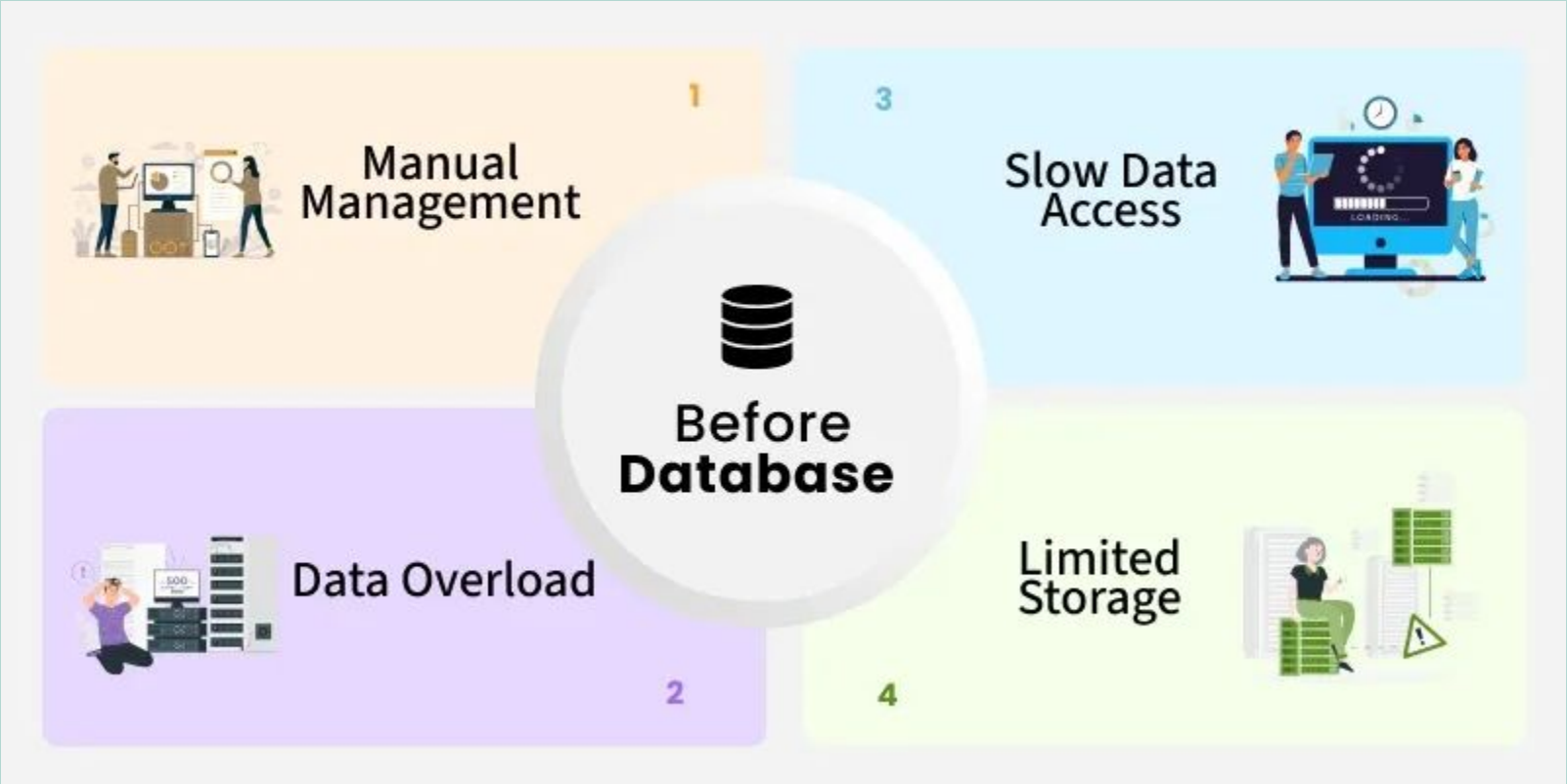
Flat File VS Database

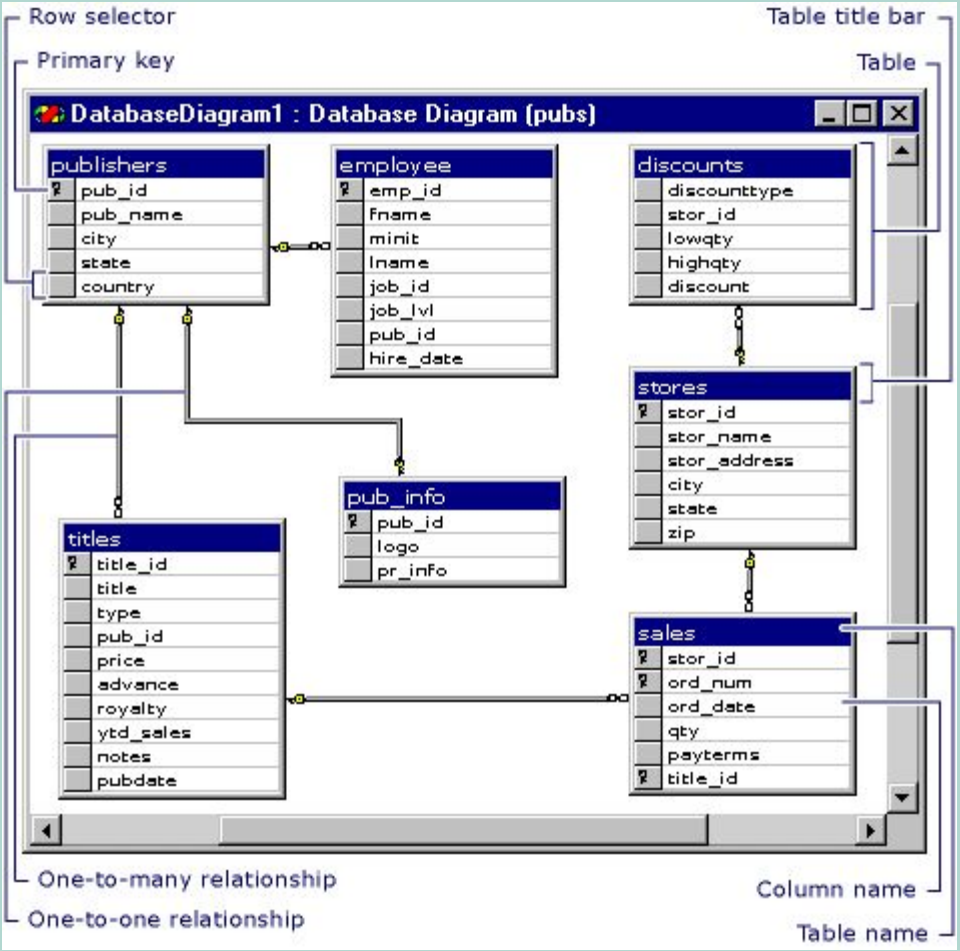
A flat file is a type of data storage that contains a single table of data, with each row representing a single record and each column representing a field within that record.

VS

A database is a collection of data that is organized and stored in a structured way. It allows for the storage, retrieval, and manipulation of data in a way that is efficient and easy to use.

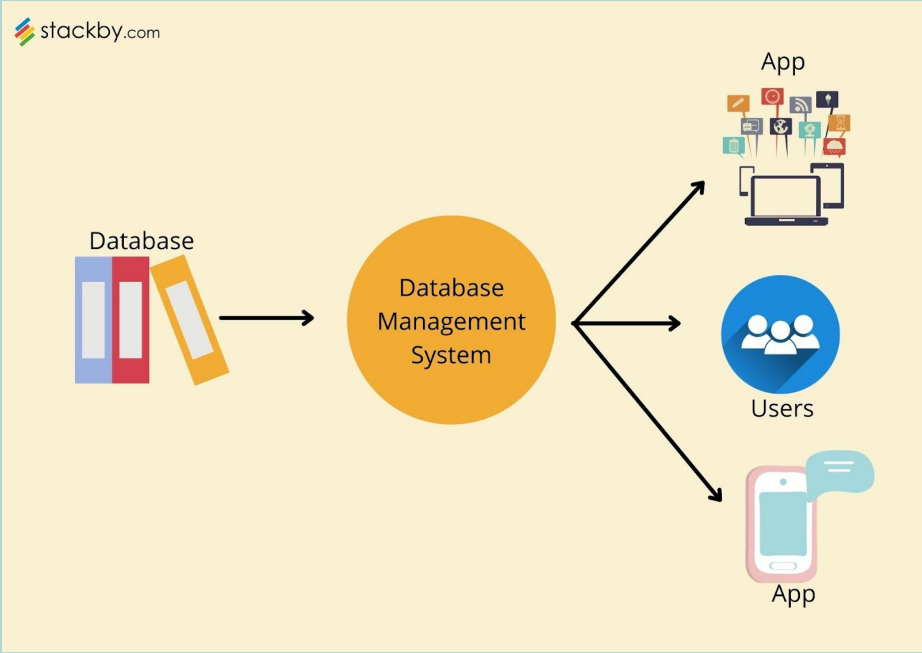
Feature	Flat File	Database
Structure	Simple, unstructured format with records stored in plain text files	Highly structured format with tables, rows, and columns
Organization	Records are stored in a single file or multiple files, but there is no relationship between them	Records are organized into tables, with relationships established between tables through keys and indexes
Access	Data is accessed by reading the entire file sequentially, making it less efficient for large datasets	Data is accessed using SQL (Structured Query Language) commands, which are more efficient for large datasets and allow for more advanced querying and manipulation
Scalability	Limited scalability, as adding more data requires creating new files or appending to existing ones	High scalability, as data can be easily added, updated, and deleted without affecting the overall structure of the database
Security	Limited built-in security features, and data is vulnerable to corruption or loss if the file is not backed up properly	Built-in security features, such as user accounts and permissions, and data is protected against corruption or loss through automatic backups and transaction logs
Examples	CSV, TSV files	SQL databases, like MySQL, Oracle, SQL Server

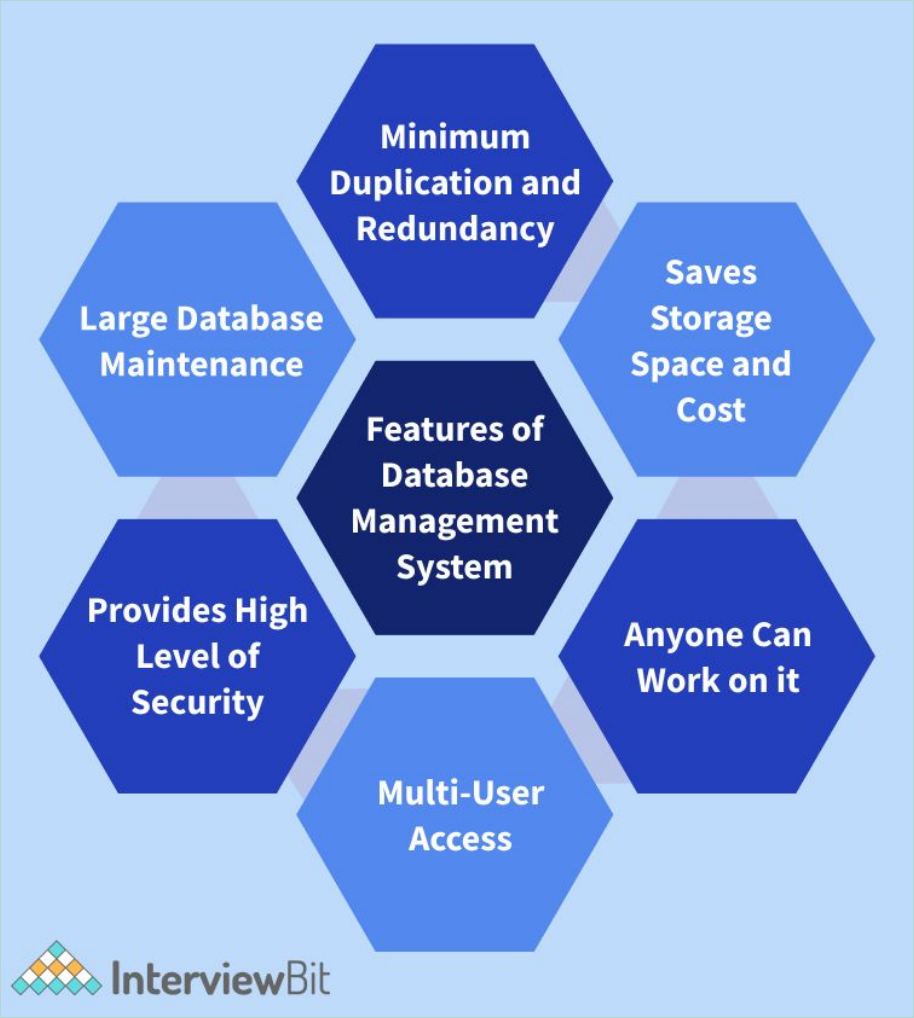

1
Manual Management
3
Slow Data Access
Before Database
Data Overload
2
Limited Storage
4


Row selector
Table title bar
Primary key
Table
DatabaseDiagram1 : Database Diagram (pubs)
publishers
pub_id
pub_name
city
state
country
employee
emp_id
fname
minit
lname
job_id
job_lvl
pub_id
hire_date
discounts
discounttype
stor_id
lowqty
highqty
discount
stores
stor_id
stor_name
stor_address
city
state
zip
pub_info
pub_id
logo
pr_info
titles
title_id
title
type
pub_id
price
advance
royalty
ytd_sales
notes
pubdate
sales
stor_id
ord_num
ord_date
qty
payterms
title_id
One-to-many relationship
Column name
One-to-one relationship
Table name

DBMS

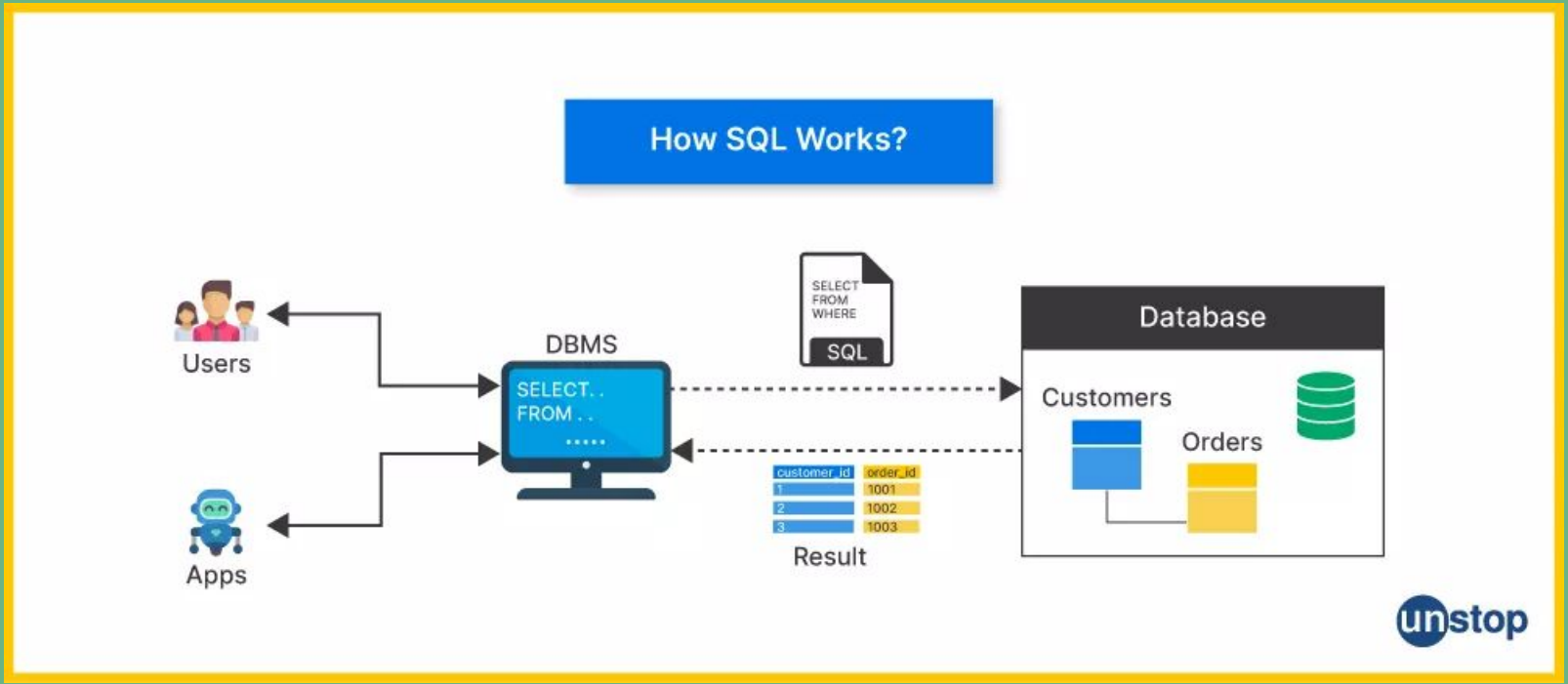
- A Database Management System (DBMS) is **software that allows users and applications to add, organize, and manage data** in databases
- **Why it's important:** It enables efficient querying, updates, and control over large datasets
- **Big picture:** A DBMS acts as the interface between users/ applications and the data, enabling reliable storage and retrieval at scale




Minimum Duplication and Redundancy
Large Database Maintenance
Saves Storage Space and Cost
Features of Database Management System
Provides High Level of Security
Anyone Can Work on it
Multi-User Access
InterviewBit

DBMS Examples

SQL



SQL is a **language used to query and manipulate data** in a database

Essential SQL Components

Tables

Tables store and structure data, organizing it into columns and rows efficiently.

Statement Commands

SQL statements create, modify, and secure database and allow access to work with data

Queries

Queries retrieve and manipulate data, answering complex questions.

Stored Procedures

Stored procedures perform routine database operations, improving efficiency and reliability.

SQL COMMANDS

SQL Command Types

- **DDL** — Data Define Language
  - Create
  - Alter
  - Drop
  - Truncate
  - Rename
- **DML** — Data Manipulation Language
  - Insert
  - Update
  - Delete
  - Merge
- **DCL** — Data Control Language
  - Grant
  - Revoke
- **TCL** — Transaction Control Language
  - Commit
  - Rollback
  - Save Point
- **DQL** — Data Query Language
  - Select

Techie Programmer

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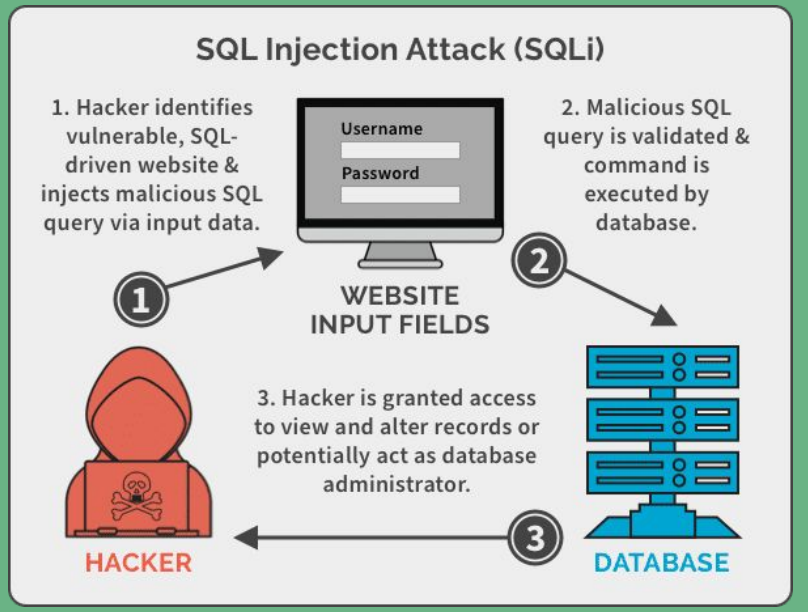
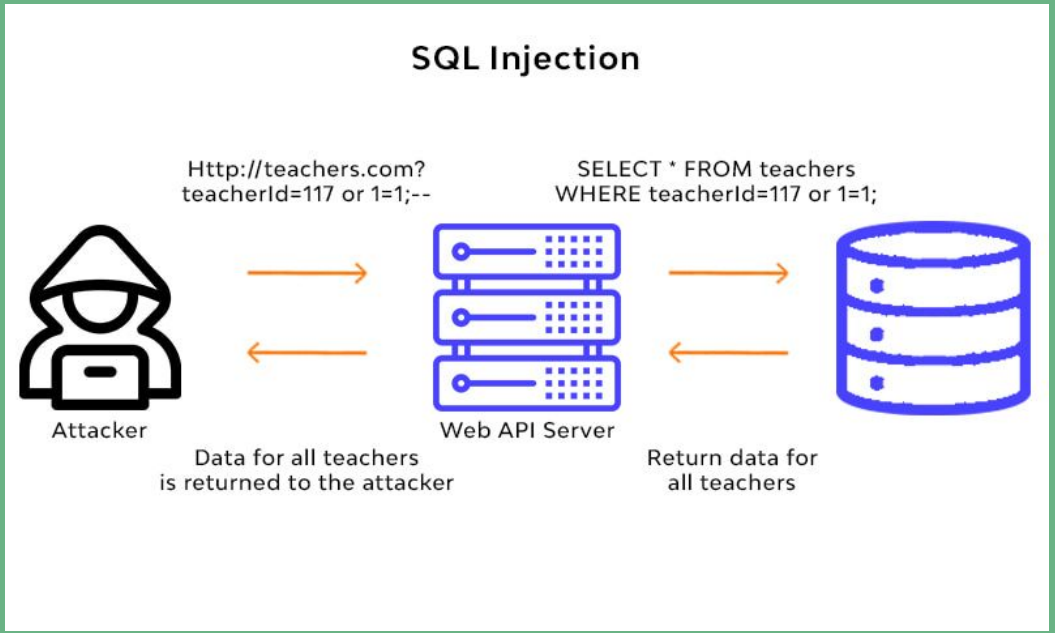
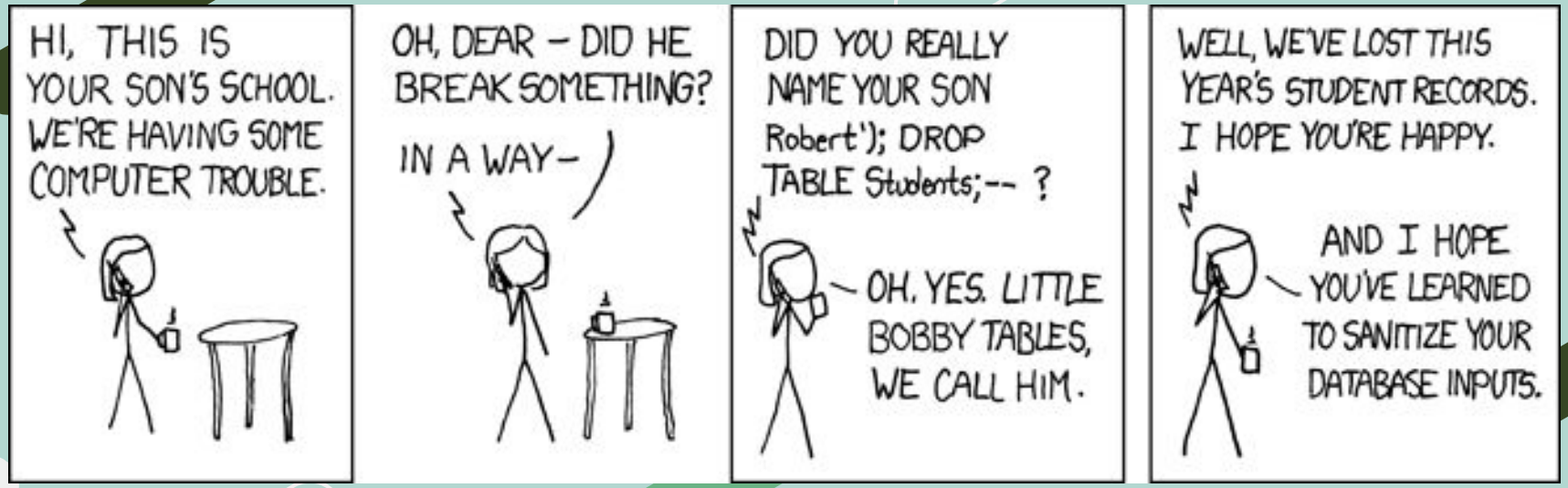
SELECT name, MIN(year) AS oldest_work_from FROM artworks JOIN artists ON artworks.artist_id = artists.id WHERE title != “The Mona Lisa” GROUP BY name HAVING MIN(year) < 1700 ORDER BY MIN(year);

To choose columns	SELECT	name, MIN(year) AS oldest_work_from
To choose a table	FROM	artworks
To join a table	JOIN	artists
	ON	artworks.artist_id = artists.id
To filter records	WHERE	title != “The Mona Lisa”
To group records	GROUP BY	name
To filter groups	HAVING	MIN (year) < 1700
To sort output	ORDER BY	MIN(year);

BONUS: SQL Injection

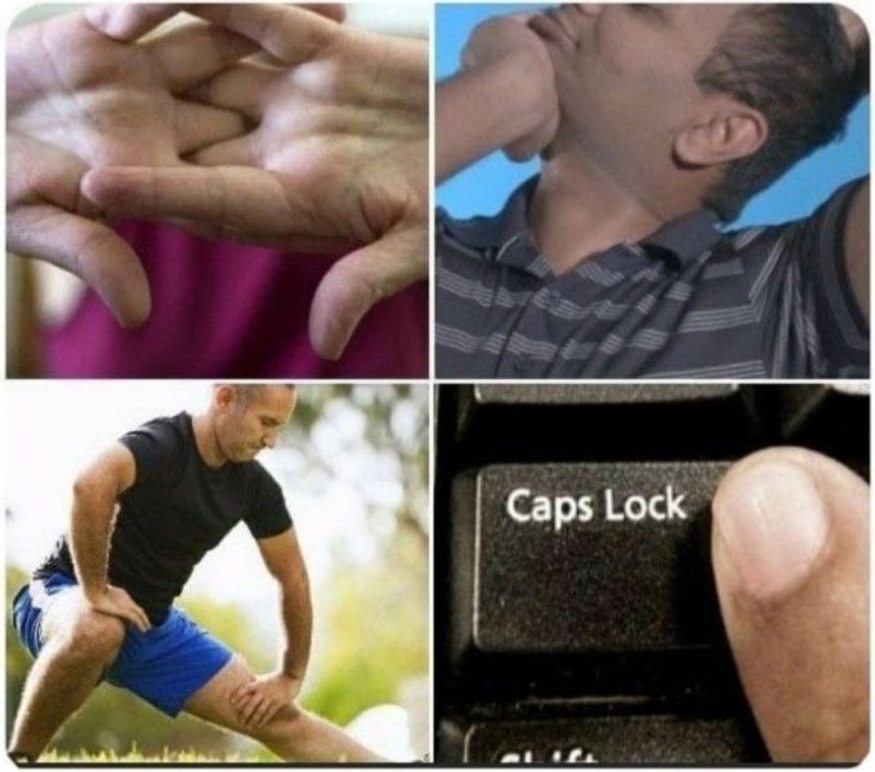





HI, THIS IS YOUR SON'S SCHOOL. WE'RE HAVING SOME COMPUTER TROUBLE.
OH, DEAR – DID HE BREAK SOMETHING?
IN A WAY–
DID YOU REALLY NAME YOUR SON Robert'); DROP TABLE Students;-- ?
OH. YES. LITTLE BOBBY TABLES, WE CALL HIM.
WELL, WE'VE LOST THIS YEAR'S STUDENT RECORDS. I HOPE YOU'RE HAPPY.
AND I HOPE YOU'VE LEARNED TO SANITIZE YOUR DATABASE INPUTS.

Karen Kringle
@KarenMN
Follow
He's making a database
He's sorting it twice
SELECT * from contacts WHERE behavior = 'nice'
SQL Clause is coming to town
RETWEETS
3,960
LIKES
2,920
5:02 AM - 16 Dec 2015

SQL programmers be like



EXERCISE

Lfg